Shu Wei Chou Chen

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Centro de Investigación en Matemática Pura y Aplicada (Research Center in Pure and Applied Mathematics – CIMPA),
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EDUCATION

2016 – 2020 **Ph.D. in Statistics.** University of São Paulo (São Paulo, Brazil).
Thesis title: *Locally Stationary Processes with Stable and Tempered Stable Innovation.*
Advisor: Prof. Dr. Pedro A. Morettin.

2012 – 2015 **M.Sc. in Statistics.** University of Costa Rica (San José, Costa Rica).
Thesis title: *Un modelo jerárquico Bayesiano espacio-tepmoral con variable respuesta de Poisson: aplicación al caso de VIH/SIDA como enfermedad rara en Costa Rica*.
Advisor: Prof. Dr. Ricardo Alvarado.

2008 – 2012 **BS. in Statistics.** University of Costa Rica (San José, Costa Rica).

PROFESSIONAL EXPERIENCE

March 2025 – present **Full Professor.** School of Statistics, University of Costa Rica (UCR), San José, Costa Rica.

March 2026 – present **Director.** Master's Program in Statistics, UCR.

2024- February 2026 **Subdirector.** Master's Program in Statistics, UCR.

November 2023 – March 2025 **Associate Profesor.** School of Statistics, UCR.

March 2020 – present **Researcher.** Research Center in Pure and Applied Mathematics (CIMPA), UCR.

March 2022 – November 2023 **Assistant Profesor.** School of Statistics, UCR.

March 2020 – March 2022 **Invited Profesor.** School of Statistics, UCR.

2012 – 2015 **Lecturer.** School of Statistics, UCR.

TEACHING EXPERIENCE

March 2020 – present *School of Statistics, University of Costa Rica.*

- *Undergraduate level in Statistics:* Statistical Theory, Professional Practice I and II, Time Series Analysis.
- *Graduate level in Statistics and Mathematics:* Time Series Analysis, Advanced topics in Time Series, Time Series I and II.

February 2017 – December 2019 — *Institute of Mathematics and Statistics, University of São Paulo.* Assistant for the courses of:

- *Statistics for other majors:* Introduction to Probability and Statistics.
- *Graduate level in Statistics:* Time Series, Spectral Analysis of Time Series, Financial Time Series Modeling.

March 2012 – March 2015 — *School of Statistics, University of Costa Rica.*

- *Statistics for other majors:* Economic, Social and Biological Sciences.
- *Undergraduate level:* Probability and Statistics Theory.
- *Graduate level in Statistics:* Experimental Design, Regressions, and Statistical Methods.

EXPERIENCE IN RESEARCH PROJECTS

As Principal Investigator:

2024-present — Análisis espacio-temporal del impacto de factores climáticos y de contaminación sobre las hospitalizaciones respiratorias. *School of Statistics, University of Costa Rica (UCR).*

2023-2026 — Desarrollo de modelos aditivos con errores simétricos autorregresivos para series temporales: generalizaciones y los métodos de pronósticos. *Research Center in Pure and Applied Mathematics (CIMPA), UCR.*

2015-2016 — Análisis espacio-temporal de VIH/SIDA en Costa Rica. *Central American Population Center (CCP), UCR.*

As Associate Researcher:

2026-present — Métodos bayesianos jerárquicos para la detección y atribución del cambio climático. *CIMPA, UCR.*

2023-2025 — Potencial aumento del riesgo en Costa Rica y Nicaragua por causa de ciclones tropicales en el Caribe. *Center for Geophysical Research (CIGEFI), UCR.*

— El costo de una vida digna en Costa Rica. Ingreso vital y la construcción de metodologías para el cálculo de variaciones subnacionales. *Institute of Social Research (IIS), UCR.*

2023-2026 — Uso de modelos de aprendizaje automático para el pronóstico de caudales extremos (mínimos y máximos). *CIMPA, UCR.*

— Análisis de extremos espacio-temporales con modelos bayesianos jerárquicos. *CIMPA, UCR.*

— El efecto de Los determinantes sociales en la propagación de enfermedades infecciosas. *CIMPA, UCR.*

2022-2023 — Use of machine learning as a predictive tool for vector-borne diseases in Costa Rica. *CIMPA, UCR.*

2020-2023 — Desarrollo e implementación eficiente de modelos para procesos espacio-temporales a través de la aproximación de resolución múltiple. *CIMPA, UCR.*

2014 — Determinación de líneas base en indicadores para evaluar el manejo integrado de recursos marino-costeros en Costa Rica. *Center for Research in Marine and Limnological Sciences (CIMAR), UCR.*

PUBLICATIONS

Journal articles:

1. M. Camacho, A. Cedeño, L. A. Barboza, **S. W. Chou-Chen**, M. J. Gómez y H. G. Hidalgo, «Hurricane Impact Index: Capturing Direct and Indirect Cyclone Effects in Central America», Earth Systems and Environment, 10.1007/s41748-025-00999-z (2026).
2. C. W. S. Chen, **S. W. Chou-Chen** y H.-H. Liao, «Spatio-temporal modelling of dengue counts in the Central Valley of Costa Rica», Epidemiology and Infection **154**, e34 (2026).
3. **S. W. Chou-Chen**, R. A. Oliveira y G. A. Paula, «Forecasting methods for additive linear models with autoregressive symmetric errors», Journal of Applied Statistics **0**, 1-28 (2026).
4. **S. W. Chou-Chen** y E. Parra-Rodríguez, «Zero-inflated and distributed lag nonlinear models with random effects for assessing environmental impacts on respiratory health in peripheral regions of Costa Rica», Frontiers in Public Health **14**, 10.3389/fpubh.2026.1753511 (2026).
5. **S. W. Chou-Chen**, A. Quesada-Román y D. Vargas-Sanabria, «Statistical Modeling of Wildfire Occurrence Based on Geomorphometry in La Amistad International Park's Buffer Zone», Environmental Modeling & Assessment, 10.1007/s10666-026-10100-1 (2026).
6. H. G. Hidalgo, S. W. Chou-Chen, K. A. McKinnon, S. Pascale, D. Quesada-Chacón, E. J. Alfaro, P. Bautista-Solís, P. M. Pérez-Briceño, H. F. Diaz, T. Maldonado, E. R. Rivera y T. Nakaegawa, «Detection and attribution of trends of meteorological extremes in Central America», Climatic Change **178**, 95 (2025).
7. **S. W. Chou-Chen** y L. A. Barboza, «Forecasting hospital discharges for respiratory conditions in Costa Rica using climate and pollution data», Mathematical Biosciences and Engineering **21**, 6539-6558 (2024).
8. **S. W. Chou-Chen**, R. A. Oliveira, I. Raicher y G. A. Paula, «Additive partial linear models with autoregressive symmetric errors and its application to the hospitalizations for respiratory diseases», Statistical Papers **65**, 5145-5166 (2024).
9. L. A. Barboza, **S. W. Chou-Chen**, M. Alfaro Córdoba, E. J. Alfaro y H. G. Hidalgo, «Spatio-temporal downscaling emulator for regional climate models», Environmetrics **34**, e2815 (2023).
10. L. A. Barboza, **S. W. Chou-Chen**, P. Vásquez, Y. E. García, J. G. Calvo, H. G. Hidalgo y F. Sanchez, «Assessing dengue fever risk in Costa Rica by using climate variables and machine learning techniques», PLOS Neglected Tropical Diseases **17**, 1-13 (2023).
11. **S. W. Chou-Chen**, L. A. Barboza, P. Vásquez, Y. E. García, J. G. Calvo, H. G. Hidalgo y F. Sanchez, «Bayesian spatio-temporal model with INLA for dengue fever risk prediction in Costa Rica», Environmental and Ecological Statistics **30**, 687-713 (2023).
12. **S. W. Chou-Chen** y P. A. Morettin, «A two-step estimation procedure for locally stationary ARMA processes with tempered stable innovations», Brazilian Journal of Probability and Statistics **37**, 155-176 (2023).
13. Y. E. García, **S. W. Chou-Chen**, L. A. Barboza, M. L. Daza–Torres, J. C. Montesinos-López, P. Vásquez, J. G. Calvo, M. Nuño y F. Sanchez, «Common patterns between dengue cases, climate, and local environmental variables in Costa Rica: A wavelet approach», PLOS Global Public Health **3**, 1-22 (2023).
14. S. Gómez-Campos, M. Cubero-Corella, M. F. Salas García y **S. W. Chou-Chen**, «Patrones de movilidad en tiempos de COVID-19: un enfoque con big data», Epi-SCIENCE **1**, 10.15517/es.2023.55398 (2023).
15. P. Vásquez, F. Sanchez, L. A. Barboza, Y. E. García, J. G. Calvo, **S. W. Chou-Chen** y G. Mery, «Mathematical and statistical models for the control of mosquito-borne diseases: the experience of Costa Rica», Rev Panam Salud Publica **113**, https://doi.org/10.26633/RPSP.2022.113 (2022).

16. **S. W. Chou-Chen** y P. A. Morettin, «Indirect inference for locally stationary ARMA processes with stable innovations», Journal of Statistical Computation and Simulation **90**, 3106-3134 (2020).

17. **S. W. Chou-Chen** y R. Alvarado-Barrantes, «Un Modelo jerárquico Bayesiano espacio-temporal con variable de conteos: aplicación de VIH/SIDA en Costa Rica.», Comunicaciones En Estadística **11**, 9-35 (2018).

18. **S. W. Chou** y E. E. Gutiérrez-Espeleta, «Ecuación para estimar la biomasa arbórea en los bosques tropicales de Costa Rica.», Revista Tecnología En Marcha **26**, 41-54 (2013).

Conference Proceedings

1. J. P. Arroyo-Castro y **S. W. Chou-Chen**, «Unsupervised Detection of Anomaly in Public Procurement Processes», en Data Science, Classification, and Artificial Intelligence for Modeling Decision Making, ed. por J. Trejos, T. Chadjipadelis, A. Grané y M. Villalobos (2025), págs. 23-32.

2. **S. W. Chou-Chen** y O. Hernandez, «Teaching Basic Statistics to Blind Students», en Proceedings 61th International Statistical Institute World Statistics Congress, Proceedings paper (2017).

Other publications:

1. **S. W. Chou-Chen** y O. Hernández, *Teaching basic statistics to blind students*, ISLP Newsletter of the International Statistical Literacy Project. `https://iase-web.org/sites/default/files/2025-08/ISLP-Newsletter-Vol-9.2-December-2017.pdf`, 2017.

CONFERENCES

2025
- **LXIII Mini-Congreso CIGEFI 2025.** Centro de Investigaciones Geofísicas, Uiversidad de Costa Rica. *"Análisis de datos funcionales sobre el impacto del clima y la contaminación en las hospitalizaciones por problemas respiratorios en Costa Rica."* **Conference, Contributed Talk**).
- **Seminar**. RESEARCH CENTER IN MATHEMATICS AND APPLICATIONS Research Center in Mathematics and Applications (CIMA), University of Évora, Portugal *"Additive Partial Linear Models with Symmetric Autoregressive Errors and their Forecasting Methods."* (**Seminar, Invited Talk**).

2024
- **Interim Conference of the Asian Regional Section of the International Association for Statistical Computing (IASC-ARS Interim Conference 2024).** *"Additive partial linear models with autoregressive symmetric errors and its application to the hospitalizations for respiratory diseases."* **Conference, Contributed Talk**).
- **Seminar**. Department of Statistics and Institute of Data Science, National Cheng Kung University, Tainan, Taiwan. *"Additive Partial Linear Models with Autoregressive Symmetric Errors and Its Forecast Methods."* (**Seminar, Invited Talk**).
- **Simposio Internacional sobre Métodos Matemáticos Aplicados a las Ciencias, SIMMAC XXIV**, Universidad de Costa Rica, Liberia, Costa Rica. *"Modelos lineales parciales aditivos con error simétrico autorregresivo para datos de series temporales."* (**Conference, Invited Talk**).

2023
- **XII Foro Internacional de Estadística Aplicada.** Facultad de Estadística e Informática, Universidad Veracruzana, Xalapa-Enríquez, Veracruz, México. *"Modelos predictivos aplicados a datos climáticos y dengue: un enfoque espacio-temporal y de aprendizaje de máquinas."* (**Conference, Invited Talk**).

2022
- **XV Jornadas de Análisis Estadístico**. Escuela de Estadística, UCR. *"Predicción de los riesgos de dengue en Costa Rica mediante el uso de variables climáticas y técnicas de aprendizaje automático."* (**Conference, Invited Talk**).

- **Workshop on cliamte change, human behavior, and vector-borne diseases**. Department of Public Health Sciences, University of California, Davis, California, United States. *"Assessing dengue fever risk in Costa Rica by using climate variables and machine learning techniques."* (**Workshop, Invited Talk**).

2021
- **Seminar**. Escuela de Estadística, UCR, San José, Costa Rica. *"Processos localmente estacionarios con innovaciones estables y estables temperadas."* (**Seminar, Invited Talk**).
- **Seminar**. Universidade Federal do Rio Grande do Norte, Natal, Brasil. *"Processo $\alpha-$estável localmente estacionário."* (**Seminar, Invited Talk**).
- **Seminar**. CIMPA-UCR, San José, Costa Rica. *"Proceso localmente estacionario con innovaciones estables"* (**Seminar, Invited Talk**).

2019
- **Seminar**. Universidade Federal de Santa Maria, Santa Maria, Rio Grande do Sul, Brasil. *"Processo $\alpha-$estável localmente estacionário."* (**Seminar, Invited Talk**).
- **18th ESTE - Time Series and Econometrics Meeting**. Gramado, Rio Grande do Sul, Brasil. *"Locally stationary ARMA processes with tempered stable innovations."* (**Conference, Contributed Talk**).
- **XVI EMR - Escola de Modelos de Regressão.** Pirenópolis, Goiás, Brasil. *"Indirect estimation for α-stable time-varying ARMA processes."* (**Conference, Contributed Talk**).

2018
- **23º SINAPE - Simpósio Nacional de Probabilidade e Estatística**. São Pedro, São Paulo, Brasil. *"Indirect estimation for alpha-stable time-varying AR(1) processes."* (**Conference, Contributed Talk**).

2017
- **61st World Statistics Congress**. Marrakech, Marruecos. *"Teaching Basic Statistics to Blind Students."* (**Congress, Poster Presentation**).
- **XV EMR - Escola de Modelos de Regressão.** Goiania, Goiás, Brasil. *"Time series analysis on the Global Mean Sea Level."* (**Conference, Poster Presentation**).
- **XVII Escola de Séries Temporais e Econometria**. (**Conference, Passive Participation**).

2015
- **RSS 2015 Annual Conference. University of Exeter.** *"A simulation study of spatio-temporal Bayesian models with Poisson response."* (**Conference, Poster Presentation**).
- **60th World Statistics Congress.** International Statistical Institute (**Congress, Passive Participation**).

2014
- **Joint Statistical Meeting**. American Statistical Asociation (**Congress, Passive Participation**).

2013
- **Primer Simposio Centroamericano de Estadística Bayesiana**. Universidad de Costa Rica. *"Análisis espacial Bayesiano de tasas de homicidios cantonales en Costa Rica."* (**Symposium, Contributed Talk**).
- **XII Jornadas de análisis estadístico.** Universidad de Costa Rica (**Conference, Passive Participation**).

2010
- **XI Jornadas de análisis estadístico.** Universidad de Costa Rica (**Conference, Passive Participation**).

ADMINISTRATIVE AND ACADEMIC MANAGEMENT

2026-present
- Director, Master's Program in Statistics, UCR.

2024-2025
- Subdirector, Master's Program in Statistics, UCR.

2022-present
- Representative of the Graduate Program in Statistics on the Advisory Council of the Research Center for Development Observatory (CIOdD), UCR.
- Representative of the Graduate Program in Statistics on the Scientific Council of CIOdD, UCR.

2020-present
- Coordinator of the Evaluation, Guidance, and Student Affairs Committee at the School of Statistics, UCR.

- Member of the Master's Program in Statistics Committee, UCR.

2020-2022
- Coordinator of the Professional Practice in the Bachelor's Program in Statistics.

AWARDS AND HONORS

- UCREA Grant for the project: *"Potencial aumento del riesgo en Costa Rica y Nicaragua por causa de ciclones tropicales en el Caribe."* Participation as an associate researcher during 2023-2025.
- UCREA Grant for the project: *"El costo de una vida digna en Costa Rica. Ingreso vital y la construcción de metodologías para el cálculo de variaciones subnacionales."* Participation as an associate researcher during 2023-2025.
- Second place for Best PhD Thesis Award (2019-2020) from the Brazilian Statistical Association, Brazil.
- First place for the best poster titled: *"Indirect estimation for α-stable time-varying AR(1) processes"* at the 23rd SINAPE - National Symposium on Probability and Statistics in São Pedro, São Paulo, Brazil, 2019.
- Graduation with honors in the Academic Master's in Statistics, 2015. University of Costa Rica.
- Graduation with honors in the Bachelor's in Statistics, 2012. University of Costa Rica.
- Highest GPA in the Bachelor's in Statistics program, 2010. University of Costa Rica.
- Highest GPA in the Bachelor's in Statistics program, 2009. University of Costa Rica.

OTHER PROFESSIONAL ACTIVITIES

Editorial Board of Journals

2022-present.
- Member of the Editorial Board of Revista Epi-Science.

2020-present.
- Member of the Editorial Board of Revista de Matemática: Teoría y Aplicaciones.

Journal Review:

- Environmental and Ecological Statistics, Climate Risk Management, Plos One, Scientific Report, and Revista de Matematica: Teoria y Aplicaciones.

Workshops and Short Courses Taught

May 2023
- *Statistical Modeling and Machine Learning Applied to Climatic Data and Public Health*, Faculty of Statistics and Informatics, Universidad Veracruzana (10 hours)

January 2016
- *Data Mining*, University of Costa Rica (16 hours)

September 2014
- *Introduction to R*, University of Costa Rica (16 hours)

October 2015
- *Introduction to R*, University of Costa Rica (16 hours)

Research Stays and Academic Visits

December 2024
- Institute of Statistical Science, Academia Sinica, Taipei, Taiwan.
- Department of Statistics and Institute of Data Science, National Cheng Kung University, Tainan, Taiwan.

September 2014
- Department of Educational Psychology, University of Minnesota, United States. *Active Approach in Teaching Statistics* with Dr. Robert DelMas.

Courses and Certifications

2019
- IX WWlet – Workshop in Wavelets and Applications. Gramado/RS, Brazil. *Multiscale Analysis and Wavelets: Theory, Development, and Applications.*

- School of Regression Models. Pirenópolis/GO, Brazil. *Count Data Regression Models: Over- and Under-dispersion.*

2018
- 23rd National Symposium on Probability and Statistics – SINAPE. São Pedro/SP, Brazil:
  - *Binary Regression Models: Inference and Applications.*
  - *Bayesian Distributional Regression.*

2017
- XVII ESTE Brazilian School of Time Series and Econometrics, São Carlos, Brazil. *Statistical Inference for Nonstationary Time Series: Functional Data Analysis and Bootstrap perspectives.*

2014
- School of Statistics, University of Costa Rica. *Time Series with Financial Data.*
- JSM Professional Development, Joint Statistical Meeting, Boston, United States. *Hierarchical Bayesian Modeling and Analysis for Spatial Data.*

PARTICIPATION IN THESES AND GRADUATION PROJECTS

Thesis Advisor:

July 2025
- *"Modelo de la influencia de los factores climáticos y la contaminación del aire en la incidencia de enfermedades respiratorias en las regiones y subregiones climáticas de Costa Rica."* Professional Master's Degree in Mathematical Methods and Application of Emmanuelle Parra Rodríguez.

June 2025
- *"Modelos de cambio de régimen de Markov aplicados a temperatura del agua de ríos en el macizo Cerro de la Muerte, Costa Rica ."* Professional Master's Degree in Statistics of Paola Campos Arce.

August 2024
- *"Comparación de técnicas de análisis multivariado para detección de valores extremos con técnicas no supervisadas, una aplicación al caso de anomalías en contratación de bienes en Costa Rica."* Academic Master's Degree in Statistics of Jose Pablo Arroyo Castro.

May 2024
- *"Análisis de los casos de intoxicación reportados al Centro Nacional de Control de Intoxicaciones de Costa Rica durante el período de COVID-19 entre los años 2020 y 2022."* Professional Master's Degree in Statistics of Edgardo Gutiérrez Vega.

July 2022
- *"Modelo bayesiano espacio-temporal predictivo acumulado de personas fallecidas de Covid-19 por cantones en Costa Rica durante el mes de mayo de 2021."* Professional Master's Degree in Statistics of Ricardo Zamora Mennigke.

January 2022
- *"Nexo entre crecimiento económico y desarrollo del sistema financiero: Análisis de cointegración y series de tiempo para el caso de CR."* Professional Master's Degree in Statistics of Fabián Brenes Trejos.

SKILLS

Languages: Spanish (near-native), Mandarin (native), English (advanced), and Portuguese (advanced).
Statistical Programs: R, SPSS, JMP, STATA, CSPro, Eviews, Winsteps, OpenBUGS.